MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, MULTIFUNCTION, -70°C TO +150°C, SINGLE, EHU-18/A

This amendment forms a part of Military Specification MIL-I-7749A(WEP) dated 1 June 1961 and has been approved by the Bureau of Naval Weapons, Department of the Navy

Page 1, title, after "SINGLE": Add

"EHU-18/A"

Page 1, paragraph 1.1: Delete and substitute

"1.1 Scope - This specification covers certain design and all performance requirements for the procurement of a single pointer, electrical resistance *temperature indicator* designated as Type EHU-18/A."

Page 3, paragraph 4.3: Delete and substitute

"4.3 Acceptance Tests: *Acceptance tests are those tests performed on* instruments manufactured and submitted for acceptance under contract and shall consist of individual tests and Sampling Plans A and B tests. See 6.3. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

4.3.1 Individual Tests - The individual tests shall consist of the tests listed below, conducted in the order given, on each instrument:

(1) *Examination of Product*

(2) Scale Error at Room Temperature

(3) Position Error

(4) Friction Error

(5) Sealing

(6) Insulation Leakage

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4.3.2 Sampling Plans and Tests -

4.3.2.1 Sampling Plan A - Samples shall consist of three indicators selected at random by the Inspector from each lot of 100 or less which have passed the Individual Tests. A lot shall consist of identical instruments with the same manufacturer's part number, manufactured under substantially the same conditions and submitted at substantially the same time. Instruments which have been subjected to the Sampling Plan A tests shall not be delivered on contract until they have been reconditioned and again subjected to all Individual Tests. Each sample shall be plainly identified by a securely attached durable tag marked with the following information:

_____ (Name of item)

Submitted by (manufacturer's name, date) for production acceptance (Sampling Plan A) testing in accordance with Contract/Order No. _____.

4.3.2.1.1 Sampling Plan A Tests - The sampling plan A tests shall consist of those listed below, conducted in the order given, on each sample:

(1) Reduced Voltage Operation

(2) Scale Error at Low Temperature

(3) Scale Error at High Temperature

(4) Magnetic Effect

(5) Resistance Bulb Current

(6) Open Circuit and Short Circuit

(7) Vibration Error

(8) Vibration Failure

4.3.2.2 Sampling Plan B - Unless otherwise specified, the samples shall consist of three indicators selected at random by the Inspector from the first 15 instruments produced on contract. The samples shall be selected within ten days after manufacture and forwarded to the laboratory specified. (see 6.3). Samples which have been subjected to the Sampling Plan B tests shall not be delivered on contract until they have been reconditioned and again subjected to all the Individual Tests. Each sample shall be plainly identified by securely attached, durable tags marked with the following information:

_____ (Name of Item)

Submitted by (manufacturer's name, date) for production acceptance (Sampling Plan B) testing in accordance with Contract/Order No. _____.

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4.3.2.2.1 Sampling Plan B Tests - The sampling plan B tests shall consist of those listed below, conducted in the order given, on each sample:

(1) Sampling Plan A Tests

(2) Response Time

(3) Damping Factor

(4) Power Consumption

(5) Bulb Lead Resistance Effect"

Page 3, paragraph 4.6 thru 4.7.2.4: Delete and substitute

"4.6 Test Methods -

4.6.1 Individual Test Methods - The individual test methods shall be those described in Specification MIL-I-7070 as modified herein.

4.6.1.1 Scale Error at Room Temperature - The test resistance specified in table I shall be applied to the resistance bulb terminals of the indicator. The test voltage shall be applied and the scale errors at the test points specified in table I shall not exceed the tolerances specified in table I.

4.6.1.2 Position Error - When the indicator is subjected to the Position Error Test, the maximum change in indication shall not exceed 4°C. With no voltage applied to the indicator, the pointer need not be off scale.

4.6.1.3 Friction Error - When the indicator is subjected to the Friction test, the maximum difference in reading before and after tapping shall not exceed 4°C.

4.6.1.4 Insulation Leakage - When the indicator is subjected to the Insulation Leakage Test, the insulation resistance shall be a minimum of 20 megohms.

4.6.2 Sampling Plan Test Methods - The sampling plan test methods shall be those described in Specification MIL-I-7070 as modified herein.

4.6.2.1 Sampling Plan A Test Methods

4.6.2.1.1 Reduced Voltage Operation - The indicator shall be subjected to a Scale Error Room Temperature Test with an applied voltage of 22.5V at the test points specified in table I. The indications at the reduced voltage shall not differ from the indications with 28.5V applied by more than the tolerances specified in table I.

4.6.2.1.2 Magnetic Effect - When the indicator is subjected to the Magnetic Effect test, it shall not cause the magnetic compass to deflect more than 5 degrees.

4.6.2.1.3 Resistance Bulb Current - A total resistance of 100 ohms shall be applied across the bulb circuit terminals of the indicator, the test voltage applied, and the bulb circuit current measured. The current consumed in the resistance bulb circuit current measured. The current consumed in the resistance bulb circuit shall not exceed 22 ma.

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4. 6. 2. 1. 4 Vibration Error - When the indicator is subjected to the Vibration Error test, the maximum amplitude of pointer oscillation throughout the frequency range shall not exceed 4°C. The maximum difference between the original position of the pointer during vibration shall not exceed 4°C. Pointer oscillation shall not be considered as pointer variation.

4. 6. 2. 2 Sampling Plan B Test Methods -

4. 6. 2. 2. 1 Response Time - When the indicator is subjected to the Response Time test, the response time shall not exceed 3 seconds.

4. 6. 2. 2. 2 Damping Factor - When the indicator is subjected to the Damping Factor test, the damping factor shall not be less than 1. 6.

4. 6. 2. 2. 3 Power Consumption - When the indicator is subjected to the Power Consumption test, the power consumption shall not exceed 1. 5w.

4. 6. 2. 2. 4 Bulb Lead Resistance Effect - The indicator shall be connected as shown on figure 2 and subjected to a scale error at room temperature test. The indications with bulb lead resistance shall not differ from the indications without bulb lead resistance, by more than ±3°C."

Page 4: Add the following new paragraphs

"6. 3 Ordering Data - Procurement documents should specify the following:

(1) Title, Number and Date of this Specification.

(2) The type number, EHU-18/A and quantity of indicators desired.

(3) Whether Sampling Plan B is to be omitted and if not, add the name and address of the laboratory to which the samples should be submitted.

(4) Levels of packaging (see 5. 1) and packing (see 5. 2) desired.

6. 4 Drawing Requirements - The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of Specification MIL-D-70327."

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